

**PUBLISHED**

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(b)     | <p><b>Two</b> from:</p> <ul style="list-style-type: none"> <li>• An embedded system is designed to perform a dedicated/limited/single function // computer can be used to perform many different functions.</li> <li>• An embedded system has <b>dedicated</b> hardware // computer has hardware that can be used by other devices.</li> <li>• An embedded system has software that is not easily updated/reprogrammed // software can easily be updated/reprogrammed on the computer.</li> <li>• An embedded system has a microprocessor // A computer has a CPU.</li> <li>• An embedded system can be part of/built into a larger device // A computer is normally standalone</li> </ul> | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                           | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(a)     | <p><b>Three</b> from:</p> <ul style="list-style-type: none"> <li>• The <u>inference</u> engine is used ...</li> <li>• ... to <b>decide</b> which questions to ask the user ...</li> <li>• ... based on the <b>previous</b> data input.</li> <li>• Symptoms input are located in/compared to knowledge base.</li> <li>• ... then applies the rule base to the knowledge base (to decide the diagnosis)</li> </ul> | <b>3</b> |
| 5(b)(i)  | <p>Any <b>two</b> from:</p> <p>Examples:</p> <ul style="list-style-type: none"> <li>• Sensors</li> <li>• Microprocessors</li> <li>• Actuators</li> </ul>                                                                                                                                                                                                                                                         | <b>2</b> |